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Contextual Factors Affecting Evidence-Based Practice in Orthopaedic Nursing and Rehabilitation: A Mixed Methods Study

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Keywords: bladder care | context | evidence-based practice | facilitation | hip surgery | implementation | nursing | orthopaedic care

ABSTRACT

Aim: To better understand what enables evidence-based practice, this study investigated contextual factors influencing evidence-based practice in general, and in relation to the implementation of bladder-monitoring guidelines in orthopaedic care.

Design: Convergent parallel mixed method.

Methods: This study was part of a hybrid research project across 17 Swedish orthopaedic sites. The data collection (2021–2023) included interviews with orthopaedic staff and patients post-hip surgery, patient survey free-text responses, and a staff survey on organisational context. Data were analysed with deductive content analysis and descriptive statistics, later integrated using a mixed methods approach.

Results: Evidence-based practice was supported by context factors such as staff collaboration and multiprofessional engagement. Staff addressed patient safety and equality by using evidence-based guidelines, but rarely involved the patients. Orthopaedic fast-track procedures positioned patients as passive recipients, while staff voiced a call for a more person-centred context. Positive attitudes, leadership engagement, use of champions, and adequate staffing enabled evidence-based practice, though a shortage in evaluation and high staff turnover hindered its implementation.

Conclusion: Orthopaedic context is characterised by several enabling organisational context factors for evidence-based practice, although patients lacking recognition of their needs and queries justify greater focus on person-centredness and mutual information exchange. Audit and feedback are crucial for improvements, but were lacking in the orthopaedic care context.

Implications for the Profession and/or Patient Care: Assessments of efforts made to implement evidence-based practice and its outcomes should incorporate nursing care. Slimmed care processes require attention to ensure patient participation.

Impact: The orthopaedic care context is enabling for evidence-based practice, although staff are challenged by fast-track procedures with extensive information exchange and insufficient person-centredness.

Reporting Method: The Mixed Methods Reporting in Rehabilitation & Health Sciences checklist.

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1 | Introduction

The implementation of evidence into practice aims to improve the quality and effectiveness of healthcare. When seeking to bridge any gap between best and current practice, the clinical context is a core element, as it represents both hindering and facilitating factors for knowledge implementation (Coles et al. 2020). Context refers to the setting or environment in which implementation takes place, including the local setting for implementation and the extended organisational context (i.e., where the care is provided, and directly and indirectly operated, respectively, within a healthcare service) (Harvey and Kitson 2020). Despite a growing general understanding of the role of context in relation to interventions for evidence-based practice (EBP), the orthopaedic care context has earlier received limited attention. Rather, given the global burden of musculoskeletal diseases and injuries, expectations of orthopaedic nursing and interprofessional competence and skills will continue to grow (Conway et al. 2017).

2 | Background

The orthopaedic care context, and in particular procedures such as hip surgery, is associated with an elevated risk of preventable adverse events (Flikweert et al. 2018). One such issue of importance for nursing care is the risk of bladder distention associated with urinary retention, caused by, for example, pain and immobility (Agrawal et al. 2019). To detect and act appropriately, there are EBP guidelines suggested; in Sweden, The Handbook for Healthcare provides easily accessible national sources of evidence and procedures for bladder monitoring during hospital care (Joelsson-Alm and Thulin 2021). Although these have been available for over a decade, the clinical practice guidelines (CPGs) have not fully reached everyday nursing care (Winberg et al. 2024). The Onset PrevenTIon of urinary retention in Orthopaedic Nursing and rehabilitation (OPTION) study (Eldh et al. 2021) was designed in recognition of the complexity of facilitating implementation (Moore et al. 2015), addressing both the knowledge content and structure (in this case, the bladder monitoring CPGs), and contextual factors such as staff and patient needs and expectations, along with resources and organisational factors (of the orthopaedic care context). This was a hybrid project to trial a strategy to increase recognition and adherence to the EBP bladder monitoring guidelines among healthcare staff and included both an effect trial and a process evaluation (Eldh et al. 2021; Curran et al. 2022). Supported by the integrated-Promoting Action on Research Implementation in Health Services (i-PARIHS) framework, we assumed that any successful implementation was to be prompted by the fit between the clinical evidence, the facilitation strategy, and the context, with attention primarily to the local context (Harvey and Kitson 2015).

Healthcare in general is associated with several contextual factors that commonly affect knowledge implementation, such as: culture, resource access, organisational support (readiness for change), collaboration and leadership (Squires et al. 2022). Further, context elements known as barriers to EBP and knowledge implementation (here framed as CPGs) are a lack of resources, varying levels of clinician training, and organisational support. Patients expect healthcare services to be safe and of good quality, and EBP integrates the best available (research) evidence, clinical expertise, and patient preferences (Melnyk and Fineout-Overholt 2022). Overlooking the synchronising of these components can lead to gaps between what the patients rightly expect and what is delivered (Saunders et al. 2019). So far, little is known about which components of the orthopaedic context affect knowledge implementation in general, and particularly when it comes to CPGs for nursing care, including both the scope and nature of barriers and enablers.

3 | Aim

The aim of this paper was to provide an enhanced understanding of contextual factors for evidence-based practice in general, and in relation to the implementation of bladder-monitoring guidelines in orthopaedic nursing care. The research questions were: (1) How do orthopaedic staff characterise the orthopaedic context with respect to factors known to influence evidence-based practice? and (2) What general and particular contextual factors, affecting evidence-based practice in orthopaedic hip surgery care, were described by staff and patients?

4 | Methods

4.1 | Study Design

This was a mixed-methods (MM) study, using a convergent parallel design (Creswell 2022). It was conducted in four steps, described below (Figure 1), and qualitative and quantitative data were considered equally important (Gwennan Hâf 2020). This study report employed the Mixed Methods Reporting in Rehabilitation and Health Sciences standard, MMR-RHS (Tovin and Wormley 2023).

4.2 | Study Setting and Recruitment

As part of a multicentre cluster-randomised Swedish trial and process evaluation (Eldh et al. 2021), this study comprised 17 Swedish orthopaedic sites performing hip surgery. Following consent from each head of unit, patients who were ≥ 18 years and had hip surgery during one of three time intervals (January–April 2021, May–June 2022 and April–May 2023) were identified, along with all nursing and rehabilitation staff employed at the 17 sites during those intervals. Additionally, inclusion criteria for patient interviews were being able to communicate in Swedish without assistance and not having a known cognitive impairment.

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Summary

- What is already known?
  - The orthopaedic context often represents swift procedures and multiprofessional teams, including nursing care.
- What this paper adds?
  - What enables evidence-based practice in the orthopaedic context is challenged by extensive information processing for patients in fast-track procedures. A further inclusion of patients' perspectives is required for more person-centred care, along with audit and feedback to improve service quality.
- Implications for practice/policy
  - Staff and patients' perspectives can enhance healthcare providers' provision of evidence-based practice.
- What does this paper contribute to the wider global clinical community?
  - Mixing qualitative and quantitative data provides for an enhanced understanding of contextual factors affecting evidence-based practice in orthopaedic care.
  - Orthopaedic care represents an overall enabling organisational context, though challenged by fast-track processes and staff shortages.
  - Audit and feedback during implementation of evidence-based practice were highlighted as important, in addition to a further inclusion of patients' perspectives and needs.
- *Impact statement:* This research identifies contextual factors in orthopaedic nursing and rehabilitation, important for the implementation and sustainment of evidence-based practice. Opportunities for patient participation, and the need for audit and feedback on such and other improvement efforts, are vital to the healthcare teams, nursing staff, and managers.

4.3 | Data Collection

Quantitative data was collected using the Alberta Context Tool (ACT) (Estabrooks et al. 2009) (with permission), distributed to staff via first-line managers at each site (with responses sent to the researchers).

The ACT version used includes 56 items, organised into eight dimensions of the local context, allocated by means of the theoretical framework Promoting Action on Research Implementation in Health Services, and additional literature (Estabrooks et al. 2009). The ACT dimensions include: leadership; culture; evaluation; social capital; informal interactions; formal interactions; structural/electronic resources, and; organisational slack (the dimensions are further clarified in Table S1). With one of the dimension (organisational slack, that is, the cushion of actual or potential resources at a work place), divided into three (staff, time and space), the ACT comprises 10 context concepts. The items are reported either with a Likert-scale ranging from 'strongly disagree' to 'strongly agree' (one to five) or by frequency ('never' to 'almost always'). The ACT has been shown to be valid and reliable (Squires et al. 2015), and has been translated and validated for Swedish nursing care (Eldh et al. 2013).

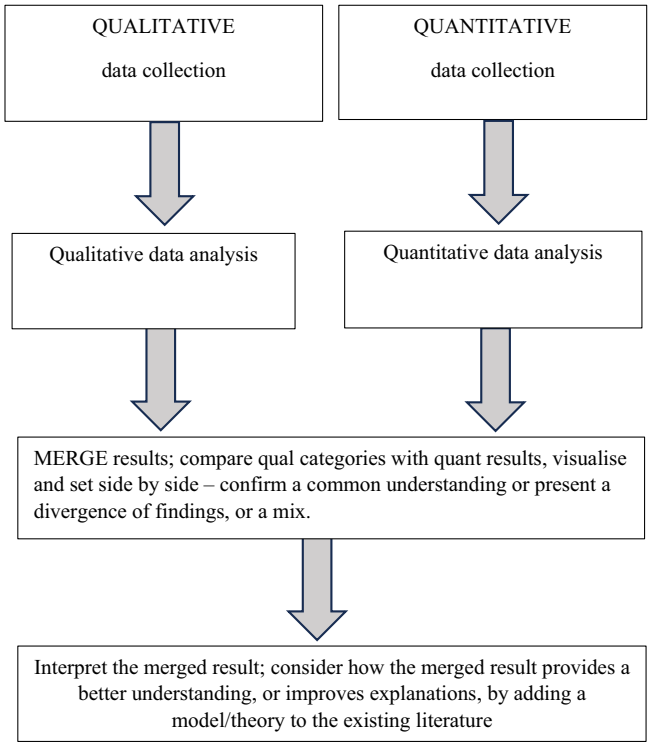


FIGURE 1 | Overview of the four phases of a mixed-methods, convergent parallel design (Gwennan Hâf 2020).

Qualitative data was collected by semi-structured interviews, and particular survey data (as below). For the individual interviews with staff and patients, respectively, a purposeful sample was drawn considering heterogeneity across sites. In addition, for the intervention sites of the trial, there were local implementation teams (consisting of nursing and rehabilitation staff and their first-line managers) which were interviewed as separate groups. All in all, the qualitative sample was appraised for information power, indicating adequacy, variability, and quality for the study (Malterud et al. 2016).

The qualitative survey data were the free-text responses from a study-specific patient survey on bladder care, where the patients were asked to recount any further bladder experiences following their orthopaedic care for hip surgery (next to four items with fixed responses on bladder care and bladder issues; these are reported elsewhere). The patient survey was sent via post to all identified patients at the 17 sites and returned by consenting patients in a pre-paid envelope to the research team. The survey data was registered in IBM Statistical Package for the Social Sciences (SPSS), version 29.0, and the verbatim transcripts were later exported to NVivo, version 14.

The authors MW and EF (PhD students; registered occupational therapist, registered nurse) conducted the group interviews and most individual interviews; some individual interviews were conducted by authors MHN (registered

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nurse, senior researcher) and ACE (registered nurse, senior researcher), and two trained researchers (registered nurses). Representing both modest and extensive experience in the clinical field (orthopaedic care), competence in knowledge implementation, academic level of context measures, and in the research methodologies applied, the research team made sure that the interview technique and format aligned prior to the interviews.

Once informed consent was obtained, the researcher assigned to interview contacted the informant(s) and agreed on a date and time. Individual staff and patient interviews were conducted via telephone for convenience and flexibility. The implementation team interviews were conducted at each site post-intervention (except one via Zoom) and digitally at the one-year follow-up. This was also per the participants' preferences. All interviews were audio recorded and transcribed verbatim prior to analysis.

For an overview of the data collection and details of this study, see Table 1, including respondent details in Table S1.

4.4 | Quantitative Data Analysis

All ACT data was registered in SPSS, and a random sample of 5% was controlled, showing full accuracy.

Descriptive analyses were conducted for demographic variables, and calculations of the ACT scores in each concept were performed according to the ACT manual (Squires et al. 2015). The predetermined cutoff scores for the concepts were set at: ≥3.0 (enabler) and <3.0 (barrier), except for Formal interaction (enabler ≥2.0, barrier <2.0), Informal interaction (enabler ≥4.5, barrier <4.5) and Structured and electronic resources (enabler ≥5.0, barrier <5.0), due to the nature of these questions (quantifying *how often* something occurred in the clinical setting) (Translating Research in Elder Care 2024). According to a prior validation study (Squires et al. 2015), these cutoff scores were set by the research group, guided by earlier literature (Fry et al. 2020).

4.5 | Qualitative Data Analysis

The qualitative data were organised as two data sets: (1) staff and local implementation group interviews, and (2) patient interviews and free-text responses. Both data sets were analysed using deductive content analysis (Elo and Kyngäs 2008). NVivo software (version 14) was used to manage the data. The analysis matrix assembled the eight dimensions of the Alberta Context Tool (ACT), considering any additional features. During analysis, each data set was subjected to three phases (Elo and Kyngäs 2008) (prior to the MM):

1. Familiarisation,
2. Organisation, by deductive coding, and
3. Abstraction, through condensed narratives influenced by phenomenological hermeneutics (Lindseth and Norberg 2004).

TABLE 1 | Qualitative and quantitative data collection and samples, procedures, and contents.

Data collection time point (year)	Qualitative data	Content	Quantitative data	Content
Baseline, T00 (2021)	Individual telephone interviews, digitally recorded; staff ($n=20$), patients ($n=32$) + Free-text responses from patient survey ($n=122$)	Individual experiences Insights of EBP in the orthopaedic context, patient experiences	ACT staff ($n=264$)	Evaluation of organisational context, aggregated scores, mapping enablers/barriers for EBP
Post intervention, T01 (2022)	Group interviews with local implementation teams performed at intervention sites, (except one via Zoom), digitally recorded ($n=7$) + Free-text responses from patient survey ($n=113$)	Group/individual experiences Insights of EBP in the orthopaedic context, patient experiences	ACT staff ($n=227$)	Evaluation of organisational context, aggregated scores, mapping enablers/barriers for EBP
One-year follow up, T02 (2023)	Group interviews with local implementation teams via Zoom, digitally recorded ($n=7$) + Free-text responses from patient survey ($n=101$)	Group/individual experiences Insights of EBP in the orthopaedic context, patient experiences	ACT staff ($n=189$)	Evaluation of organisational context, aggregated scores, mapping enablers/barriers for EBP

Abbreviations: ACT, Alberta Context Tool; EBP, evidence-based practice.

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4.6 | Mixed Methods Analysis

To conclude, an integrated MM analysis (Creswell 2022; Gwennan Hâf 2020) was conducted:

1. Quantitative and qualitative findings were revisited, and
2. These findings were integrated, illustrating findings and particularly pointing out their correspondences and differences.

4.7 | Reflexivity and Trustworthiness

The researchers conducting this study consisted of senior and junior scholars, all women, with different professional backgrounds (registered nurses, registered occupational therapist, registered physiotherapist) and academic degrees (PhD students, associate professors and full professor). The competencies of the team included clinical expertise in orthopaedic care, implementation, leadership, context measures, bladder care and rehabilitation, as well as a variety of methodological competencies (qualitative, quantitative and mixed methods). The variation in preunderstandings and positions was discussed in the research team to avoid bias during the collection, analysis and interpretation of data.

The concepts of Lincoln and Guba were applied to achieve trustworthiness (Lincoln and Guba 1986). To increase the credibility of this study, the analyses were performed by the first author in repeated dialogue with the research team, until agreement was reached on procedures, content, and outcomes. Triangulation of the experiences of the orthopaedic context was performed by addressing the narratives of both staff and patients. Transferability was aided by the heterogeneity of staff and patient samples, representing different types of hospitals (across a fairly large number of orthopaedic sites, $n = 17$), and the provision of details on the samples and settings. Confirmability was augmented by researcher reflexivity and triangulation of data sources, and analyses, including the employment of the theoretical framework. Further, the dependability was also sufficed by the triangulation of data sources.

4.8 | Ethical Considerations

Participants received written information about the study prior to giving consent, and were informed that participation was voluntary, with the option to withdraw at any time. Confidentiality was guaranteed. The OPTION trial was approved by the Swedish Ethical Review Authority, ID 2020-06140. Amendments were made: 2021-01710, 2021-02434, 2021-03755, 2021-05341-02, and 2022-02677-02. The OPTION project is registered in the US National Institutes of Health Clinical Trials Registry, ID NCT 04700969.

5 | Results

The findings are reported as originating from the quantitative and qualitative strands, respectively, with the associated ACT dimensions and additional context elements marked in bold in running text, and to conclude, the mixed methods findings. The overall integration is envisioned in a joint display table (Table S3).

5.1 | Quantitative Results and Characteristics of Participants

The ACT sample ($n = 680$) represented staff from all 17 sites (regional, local, and university hospitals), with a response rate varying from 26% to 35% (Table 2 and Table S2).

In general, the orthopaedic context dimensions were often classified as enabling EBP at all timepoints, apart from *Formal interaction* and *Informal interaction*, respectively, and *Structural/electronic resources*, reflecting barriers (Table 3).

5.2 | Qualitative Results and Characteristics of Participants

The qualitative data represented the voices of staff and patients across the 17 sites (regional, local, and university hospitals). Further details in Table S1 and Table 4.

In total, 336 free-text responses from the patient survey were added (baseline, $n = 122$; T01, $n = 113$, T02, $n = 101$). At baseline, 58.2% of the respondents were women ($n = 71$) and 41.8% men ($n = 51$), with a median age of 75 years (IQR 70–79). At the post intervention follow up (T01), 57.5% were women ($n = 65$) and 42.5% men ($n = 48$), with a median age of 75 years (IQR 71–80). And lastly, at the one-year follow up, 58.4% were women ($n = 59$) and 41.6% men ($n = 42$), with a median age of 76 years (IQR 68–84).

5.2.1 | Staff Experiences of the Orthopaedic Context

Staff described their working environment as friendly *teamwork*, where nursing and rehabilitation were openly discussed. While guidelines should preferably engage everyone, both on the local team and across the organisation, having different managers counteracted cooperation. Regular *meetings*, both formal and brief ones, facilitated the sharing of information and discussing care. Permanent and experienced staff were described as anchoring quality procedures and mentoring new staff on guidelines. The heavy *workload* and *fast-track* procedures required prioritisation, limiting individualised patient care plans. Staffing shortages were particularly problematic during weekends, potentially reducing quality of care. Staff highlighted the extensive information given to patients throughout their care trajectory and acknowledged bladder care interventions as both important and common.

Managers were considered responsible for guidelines and their implementation, and to encourage recognition and adherence through reminders and training. *Successful implementation* of EBP was described as dependent upon the managers' knowledge, feedback, and support to staff, along with its patient benefits. Staff also stressed that their own *attitudes* influenced

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TABLE 2 | Characteristics of the Alberta Context Tool (ACT) respondents, by data collection time point.

Characteristic	Baseline	T01[a]	T02[b]
Response, *n* (%)	264 (35)	227 (31)	189 (26)
Female sex, *n* (%)	238 (90)	205 (90.3)	171 (90.5)
Missing	—	*2 (0.8)*	—
Experience in orthopaedic care, years, median (IQR)	4 (1–10)	4 (1.5–9.5)	4 (1.5–9.5)
Missing, (%)	—	*2 (0.9)*	—
Experience in profession (main role), years, median (IQR)	8 (3–19)	7.3 (2.5–19.5)	8 (3–16.9)
Missing, (%)	—	*2 (0.9)*	*1 (0.5)*
Profession (main role), *n* (%)			
Registered nurse	126 (47.7)	86 (37.9)	67 (35.4)
Licenced practice nurse	113 (42.8)	124 (54.6)	106 (56.1)
Occupational therapist	10 (3.8)	1 (0.4)	2 (1.1)
Physiotherapist	12 (4.5)	7 (3.1)	5 (2.6)
Other	1 (0.4)	7 (3.1)	8 (4.2)
Missing	*2 (0.8)*	*2 (0.9)*	*1 (0.6)*
Age range, years, *n* (%)			
< 30	60 (22.7)	53 (23.3)	43 (22.8)
30–49	116 (43.9)	104 (39.4)	89 (47.1)
≥ 50	88 (33.4)	70 (26.5)	57 (30.1)
Missing	—	—	—

Abbreviation: IQR = interquartile range.
[a]Post intervention.
[b]One-year follow up.

implementation efforts, and suggested staff with a positive attitude facilitated implementation.

Adherence to EBP varied, as a result of locally adapted guidelines and inconsistent instructions. Conflicting guidelines caused confusion, but *feedback*—such as adverse event reports—provided a further understanding of EBP adherence. Regular reviews of patient records or *national quality registries* enabled *benchmarking* with other units, thus identifying local strengths and weaknesses.

Internal evidence-based guidelines were digital, and newly employed staff were briefed on the *information system*. Time scheduled for keeping up to date with EBP is needed, but this is challenged by *staffing* constraints. *Champions* like care developers and research coordinators generally support EBP and quality improvements, much like staff dedicated to supporting certain aspects of nursing, enabling others to learn.

However, *EBP* was hindered by staffs' varied reactions to change, habits, and shift work. Implementing new evidence requires *time* and proper evaluation, but this is not regularly planned or performed. Besides the lack of consistency of EBP across departments, there was a constant flow of new or revised guidelines and these, combined with time constraints and few opportunities for inhouse training, blocked successful implementation. During times of high staff turnover, managers relied on newly appointed staff and students being more prone to adopting guidelines (such as bladder monitoring guidelines); in contrast, repeated *reorganisations* or unexpected events such as the Covid-19 *pandemic* delayed EBP implementation efforts. During the pandemic, some staff emphasised their shift in focus to prioritise Covid care, cancelling administrative and educational tasks (such as advancing EBP efforts), but rapidly adapting to, for example, altered hygiene restrictions and to caring for varying groups of patients.

5.2.2 | Patients' Experiences of the Orthopaedic Context

Being cared for when having hip surgery means meeting staff who are *content* in their roles, providing security through their expertise and functioning as a *team*. When staff seemed stressed or preoccupied, there was less time for *communication*. Weekend care required more self-management due to reduced *staffing* and limited rehabilitation resources. Hip surgery care was accentuated as rapid and demanding, requiring patients to pay attention to all information shared. Unexpected events tended to stress staff, who passed this on to patients. The *information* flow hindered dialogue, because staff focused on sharing information. While *being listened to* and having one's experiences taken into consideration was important, the attentiveness of staff varied. Participating in care discussions provided a sense of control for those who preferred *to be involved*.

Bladder care was a common procedure, primarily manifesting as frequent scans, but poorly communicated; although an *understanding* of bladder issues aided patients' understanding and participation, both in hospital and after discharge, staff did not share much knowledge about bladder issues associated with hip surgery.

Patients knew of the general *economic* constraints and cutbacks in healthcare services but considered their care to be of high quality. However, the Covid-19 *pandemic* restricted visiting hours and dictated shorter in-hospital stays, which by patients was described as a deliberate action from staff to protect patients from being infected.

5.3 | Mixed Methods Findings

Both patients and staff generally depicted the orthopaedic context as a beneficial *culture*, including teamwork and a collaborative atmosphere. Adherence to fast-track procedures was

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TABLE 3 | Calculated scores for the Alberta Context Tool (ACT) concepts at baseline, T01 and T02, mapped as enablers or barriers.

	Baseline		T01[a]		T02[b]	
ACT concepts	**Median (IQR[c])**	**Enabler[d]/ Barrier[e]**	**Median (IQR)**	**Enabler[d]/ Barrier[e]**	**Median (IQR)**	**Enabler[d]/ Barrier[e]**
Leadership	**4.0** (3.6–4.3)	Enabler	**4.0** (3.7–4.6)	Enabler	**4.0** (3.7–4.6)	Enabler
Culture	**4.2** (4.0–4.5)	Enabler	**4.3** (4.0–4.7)	Enabler	**4.2** (4.0–4.5)	Enabler
Evaluation	**3.5** (3.0–4.0)	Enabler	**3.8** (3.3–4.0)	Enabler	**3.8** (3.3–4.0)	Enabler
Formal interaction[f]	**1.0** (0.0–2.0)	Barrier	**1.0** (0.3–2.0)	Barrier	**1.0** (0.0–2.0)	Barrier
Informal interaction[g]	**4.0** (2.5–5.0)	Barrier	**3.5** (2.5–5.0)	Barrier	**4.0** (3.0–5.0)	Barrier
Social capital	**4.2** (4.0–4.5)	Enabler	**4.2** (4.0–4.7)	Enabler	**4.2** (4.0–4.7)	Enabler
Structural/ electronic resources[h]	**4.0** (3.0–5.5)	Barrier	**4.5** (3.0–5.5)	Barrier	**4.5** (2.5–5.8)	Barrier
Staffing	**4.0** (3.0–4.0)	Enabler	**3.5** (2.5–4.0)	Enabler	**3.5** (2.5–4.0)	Enabler
Space	**3.7** (3.0–4.3)	Enabler	**4.0** (3.3–4.3)	Enabler	**4.0** (3.5–4.3)	Enabler
Extra time	**3.3** (2.8–3.5)	Enabler	**3.0** (2.8–3.5)	Enabler	**3.0** (2.8–3.5)	Enabler

[a]Post intervention.
[b]One-year follow up.
[c]IQR = Interquartile range.
[d]Cutoff scores (enabler ≥ 3.0, barrier < 3.0).
[e]Cutoff scores (enabler ≥ 3.0, barrier < 3.0).
[f]Cutoff score (enabler ≥ 2.0, barrier < 2.0).
[g]Cutoff score (enabler ≥ 4.5, barrier < 4.5).
[h]Cutoff score (enabler ≥ 5.0, barrier < 5.0).

mentioned by both staff and patients; this created demands on communication and rehabilitation, which could be stressful for both parties. Patients noticed staff stress, described by staff as a heavy workload. Patients suggested that this has an impact on mutual staff–patient dialogue: the provision of information, along with any opportunities to become involved and to learn of plans, was mainly dictated by the staff and affected by their stress and/or workload. While mapped as an enabling element for EBP in the survey by staff, fellow staff were noted as facilitating only if and when they showed positive attitudes and interest in implementing guidelines.

Leadership emerged as engagement and feedback, and was only noted by staff. Staff described managers as mainly responsible for EBP efforts; leadership was generally noted as enabling within the orthopaedic context.

Evaluation was not commonly performed in terms of adherence to EBP guidelines, although it was suggested as important feedback by staff. Rather, regular evaluations focused on patient satisfaction or outcomes, according to both patients and staff.

In terms of both *formal* and *informal interactions* in the orthopaedic context, patients noted how staff performed and communicated care in a confident manner, enhancing trust. Staff focused on the exchange of information with peers, highlighting educational aspects of staff or unit interactions, which were enabled by the *social capital* of the multiprofessional team or experienced colleagues. However, such conditions were seldom provided, which hindered EBP. This also related to communication in orthopaedic care, with patients depicting a transmission of information rather than a mutual dialogue—in parallel with staff wanting more time to spend with patients in order to provide better conditions for communication and patient engagement. Both staff and patients described the EBP guidelines for bladder care as seemingly important, but patients experienced a lack of information about why procedures were performed, while staff did not address involving patients or explaining why bladder care is important.

Rather, staff described *staffing* and *time* as affecting the quality of care. Staffing, *space* (i.e., premises for nursing and administrative activities), and extra time, also called *organisational slack*, were quantified as enabling EBP, although these were depicted as hindering factors in the qualitative narratives of staff. Staff turnover was described by staff as problematic, but newly employed staff seemed to reach out for guidelines, enhancing their learning process and providing safe care and rehabilitation. Thus, from an EBP perspective, guidelines and routine adherence were not necessarily hindered by the appointment of new staff.

EBP was disseminated in different ways, and staff were informed about new guidelines via *structured and electronic resources* such as digital systems, or via a constant flow of emails. This was described as challenging to keep up with and, while

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TABLE 4 | Demographic characteristics of informants in individual and group interviews.

	Baseline		T01[a]	T02[b]
	Individual interviews with staff	Individual interviews with patients	Group interviews with local implementation teams ($n=7$)	Group interviews with local implementation teams ($n=6$)
Included respondents, *n*	20	32	20	14
Length of interviews, minutes, median (IQR)	43 (36–52)	21 (15–30)	49 (40–68)	30 (23–42)
Age, years, median (IQR)	37 (28–55)	74 (65–77)	49 (41–56)	50 (42–57)
Sex				
Female, *n* (%)	15 (75)	20 (62.5)	18 (90)	13 (93)
Profession (main function), *n* (%)		N/A		
Registered nurse	12 (60)	—	5 (25)	4 (26.7)
Licenced practice nurse	5 (25)	—	6 (30)	3 (26.7)
Occupational therapist	2 (10)	—	1 (5)	—
Care coordinator	1 (5)	—	1 (5)	1 (6.7)
Manager	—	—	6 (30)	5 (33.4)
Care developer	—	—	1 (5)	1 (6.7)
Experience in orthopaedic care, years, median (IQR)	4 (1–9)	N/A	7 (3–14)	7 (4–15)

Abbreviation: IQR = interquartile range.
[a]Post intervention.
[b]One-year follow up.

meetings were more helpful, not all staff on the ward could meet and discuss issues at the same time. Furthermore, staff found *economic constraints*, *reorganisations*, and *unforeseen events* such as the pandemic slowed down EBP efforts, while patients voiced that care was seemingly satisfactory despite the suggested constraints in healthcare.

6 | Discussion

The orthopaedic context incorporated several enabling factors for EBP, but also hindering elements. Divergence in the mixed methods findings was found only for the concept organisational slack (staffing, space, time), where the ACT scores indicated this as an enabler of EBP while staff narrated this as a barrier (in the interviews). We suggest that further attention should be given to the following three context-specific aspects of knowledge implementation: the importance of evaluation and feedback on EBP efforts, along with staff–manager cooperation; the lack of patient participation in EBP; and the EBP adoption among newly employed staff. These elements are relevant to both EBP in general and to the bladder monitoring EBP specifically.

While auditing and feedback has been suggested to improve quality of care (van der Winden et al. 2023), and aid in the implementation of knowledge, staff in this study stressed that EBP efforts for nursing and allied healthcare were rarely evaluated. Evaluations are important for both staff and managers to create an awareness of quality-of-care outcomes, and to receive feedback on EBP implementation strategy outcomes. While an evaluation of the conduct of individuals and teams is warranted, it can also aid further adaptations of the local context to support and sustain EBP (Cassidy et al. 2021). Still, there is a further need for the coordinated evaluation of EBP to improve patient outcomes (Connor et al. 2023), and such evaluations could also underpin further local managerial strategies for implementing EBP.

Orthopaedic staff suggested that their managers are primarily responsible for local EBP efforts, although they also voiced a need for collaboration between staff and managers. Thus, mutual investments are required if EBP is to be implemented; an EBP-engaged, supportive, and persistent leader is known to be crucial for EBP in nursing (Mathew et al. 2024), although previous studies have shown their need for support in return for enabling such adoption (Fjordkvist et al. 2024). In the orthopaedic context, leadership was described as enabling in terms of feedback and support, but also for reminders and procuring training for staff. The use of such strategies spanning both leadership and individual levels is known to be effective (Matthew-Maich et al. 2013), although it is important that individual staff believe in the evidence-based guidelines and the gains for patients (Cassidy et al. 2021). In our study, this was exemplified by nursing staff and managers voicing that adherence to the bladder guidelines had clear benefits for hip surgery patients and could prevent adverse events.

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On the other hand, both staff and patients also described a lack of patient engagement in bladder monitoring. While staff focused on outcome measures, patients were not fully informed of its importance. This reduced the opportunities for patient participation; the lack of mutual communication depicted by patients both in this study and previously (Winberg et al. 2023) limits the recognition of individual needs, resources, and preferences. In contrast, patients who are knowledgeable about their own health can partner with staff to produce a better collaboration (Pomey et al. 2015), serving a more person-centred healthcare context (Santana et al. 2018). Nevertheless, communication constraints may lead to missed nursing care (Mandal and Seethalakshmi 2023), but these should be further addressed in terms of bladder care, to enable patients' health literacy and their engagement in self-care. The lack of communication with patients was also described as a consequence of limited time, leaving patients as recipients of information, rather than partners. Although suggested necessary to meet increasing healthcare needs in general, with no corresponding increase in resources, it hindered the staff's attention to the involvement of patients in their health and healthcare. While time (and training) are important elements of EBP, and vital to the success or failure of knowledge implementation (Alatawi et al. 2020; Alqahtani et al. 2020; Dos Portela Santos et al. 2022), time to engage with patients must be recognised and valued in nursing care, for the benefit of EBP (Thompson et al. 2008). The international code of ethics for nurses further emphasises that evidence-informed practice accompanies integrating patient preferences (International Council of Nurses (ICN) 2021). With patient perspectives being prominent, nurses are held accountable for such conduct, both in orthopaedic nursing and elsewhere. One suggested route is patient participation in guideline development (Kim et al. 2020), and also in the implementation of such knowledge.

Despite the barriers created by lack of resources such as time, staffing and staff–patient collaboration, we found the suggested willingness among newly employed orthopaedic staff and students to adopt EBP guidelines encouraging. Registered nurses with just a few years in practice have been found to be more likely to have positive beliefs towards EBP and to embed it into the organisational healthcare culture (Warren et al. 2016), providing opportunities to increase the adoption of CPGs. While it is suggested that more experienced peers with particular competencies can become facilitators by taking a lead on education and training for the benefit of increased EBP, they can both mentor and learn from their junior colleagues. Nurses with advanced nursing degrees serving as either formal or informal leaders (Warren et al. 2016) can improve EBP and also help bridge the gap in liaising with undergraduate nurses who recognise the importance of EBP in any healthcare context.

The use of the theoretical framework i-PARIHS provided guidance and structure when investigating contextual factors of importance for EBP. With the definition of context i-PARIHS suggesting a wide focus on both inner and outer context elements, our data could be well fitted into the suggested definitions. Determinant implementation frameworks, such as i-PARIHS, are suggested as useful tools to identify contextual barriers and enablers for EBP implementation (Harvey 2022), further alleviating the gap between theory and practice (Cassidy et al. 2021).

6.1 | Strengths and Limitations

The findings were reinforced by including both qualitative and quantitative perspectives, supported by the i-PARIHS framework; the complementary nature of MM minimises the limitations of quantitative and qualitative methods, respectively, while recognising the strengths of each (Gwennan Hâf 2020). The ACT enabled a broad representation of staff's measures of the organisational context, while the qualitative perspectives of interviews and free-text survey responses significantly contributed to a more detailed understanding. While the qualitative and quantitative data represented samples from regional, local, and university hospitals, this provided a heterogeneity, increasing opportunities for both generalisability and transferability (Younas and Durante 2023).

Still, as only consenting staff conducted the ACT survey, we could not further investigate whether non-respondents differed from the participants, making this a potential source of non-response bias (Toepoel and Schonlau 2017). Although the respondents were not specifically asked in the ACT to consider the conditions available for bladder guideline implementation, this was further elaborated during the interviews. Additionally, some aggregated ACT scores that are close to the cutoff values should be interpreted with caution, along with the moderate response rate of the ACT survey (26%–35%), which may introduce non-response bias and challenge the generalisability of findings (Parekh et al. 2020).

In this study, patients were involved as participants in both interviews and surveys, although future research could benefit from a closer collaboration with patient representatives to improve the design and conduct of any research project, by means of their lived experience perspectives (Baker et al. 2018).

6.2 | Recommendations for Further Research

Future research should further acknowledge combining quantitative and qualitative methods, enhanced by using a suitable theoretical framework, to enable a better understanding of how contextual factors influence EBP in healthcare. Patients' experiences must be considered in order to fully understand their perspectives on EBP guidelines and how they are aligned to improve nursing and rehabilitation outcomes.

6.3 | Implications for Policy and Practice

- The orthopaedic context in terms of evidence-based practice is better understood when combining both quantitative and qualitative data.
- The orthopaedic care context is depicted as having an overall enabling organisational context, although fast-track care processes and staff shortages create stress, challenge information exchange, and limit the time available to stay up to date.
- Nursing and rehabilitation staff in orthopaedic care should incorporate patient needs for information about evidence-based bladder monitoring.

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- Newly employed staff seeking guidance from guidelines can enable the further dissemination of EBP.
- Managers and staff should receive feedback on their EBP performance efforts incorporating nursing and rehabilitation outcomes.

7 | Conclusions

Mapping the qualitative and quantitative results, using i-PARIHS as a theoretical foundation, this mixed methods' study revealed that although orthopaedic care is challenged by fast-track processes, as well as time and resource constraints, the care context is characterised by several enabling components for EBP. Still, future EBP implementation efforts should involve patients' perspectives to enable an increased understanding among both staff and patients about EBP actions and outcomes, quality, and safety in healthcare. More recently employed staff seemed to be more dependent upon guidelines, which may enable the broader adoption of these guidelines, and their natural adherence to EBP should be valued because it facilitates implementation. Both feedback on the evaluation of EBP performance and managerial support are important factors to consider in any implementation effort in clinical nursing.

Author Contributions

All authors have agreed on the final version and meet at least one of the following criteria (recommended by the ICMJE, https://eur01.safelinks.protection.outlook.com/?url=http%3A%2F%2Fwww.icmje.org%2Frecommendations%2F&data=05%7C02%7Cmadeleine.winberg%40liu.se%7C5568c5d350904250d86c08ddc069e6fd%7C913f18ec7f264c5fa816784fe9a58edd%7C0%7C0%7C638878283264608435%7CUnknown%7CTWFpbGZsb3d8eyJFbXB0eU1hcGkiOnRydWUsIlYiOiIwLjAuMDAwMCIsIlAiOiJXaW4zMiIsIkFOIjoiTWFpbCIsIldUIjoyfQ%3D%3D%7C0%7C%7C%7C&sdata=B72GEHxbtPbu6yfc4N1GIahm7B%2Blpb%2BJa3UZOjUZXWE%3D&reserved=0): substantial contributions to conception and design, acquisition of data, or analysis and interpretation of data; drafting the article or revising it critically for important intellectual content.

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Disclosure

The authors take responsibility for the choice of statistical approach, and for the correct conduct and interpretation; the authors have adhered to the Journal's statistical guidelines. Further, authors M. Hälleberg Nyman and E. Joelsson-Alm both have extensive knowledge and experience of the descriptive statistics applied.

Ethics Statement

The OPTION trial was approved by the Swedish Ethical Review Authority, ID 2020–06140. Amendments were made: 2021-01710, 2021-02434, 2021-03755, 2021-05341-02, and 2022-02677-02.

Consent

Participants received written information about the study prior to giving consent, and were informed that participation was voluntary, with the option to withdraw at any time. Confidentiality was guaranteed.

Conflicts of Interest

The authors declare no conflicts of interest.

Data Availability Statement

The data generated and analysed during this study is not publicly available due to the protection of individual privacy but is available from the corresponding author on reasonable request.

Peer Review

The peer review history for this article is available at https://www.webofscience.com/api/gateway/wos/peer-review/10.1111/jan.70098.

References

Agrawal, K., S. Majhi, and R. Garg. 2019. "Post-Operative Urinary Retention: Review of Literature." *World Journal of Anesthesiology* 8, no. 1: 1–12. https://doi.org/10.5313/wja.v8.i1.1.

Alatawi, M., E. Aljuhani, F. Alsufiany, et al. 2020. "Barriers of Implementing Evidence-Based Practice in Nursing Profession: A Literature Review." *American Journal of Nursing Science* 9, no. 1: 35–42.

Alqahtani, N., K. M. Oh, P. Kitsantas, and M. Rodan. 2020. "Nurses' Evidence-Based Practice Knowledge, Attitudes and Implementation: A Cross-Sectional Study." *Journal of Clinical Nursing* 29, no. 1–2: 274–283. https://doi.org/10.1111/jocn.15097.

Baker, G. R., M. H. McGillion, and F. Gavin. 2018. "Engaging With Patients on Research to Inform Better Care." *Canadian Medical Association Journal* 190: S6–S8. https://doi.org/10.1503/cmaj.180816.

Cassidy, C. E., R. Flynn, and C. J. Shuman. 2021. "Preparing Nursing Contexts for Evidence-Based Practice Implementation: Where Should We Go From Here?" *Worldviews on Evidence-Based Nursing* 18, no. 2: 102–110. https://doi.org/10.1111/wvn.12487.

Coles, E., J. Anderson, M. Maxwell, et al. 2020. "The Influence of Contextual Factors on Healthcare Quality Improvement Initiatives: A Realist Review." *Systematic Reviews* 9, no. 1: 94. https://doi.org/10.1186/s13643-020-01344-3.

Connor, L., J. Dean, M. McNett, et al. 2023. "Evidence-Based Practice Improves Patient Outcomes and Healthcare System Return on Investment: Findings From a Scoping Review." *Worldviews on Evidence-Based Nursing* 20, no. 1: 6–15. https://doi.org/10.1111/wvn.12621.

Conway, D. J., R. Coughlin, A. Caldwell, and D. Shearer. 2017. "The Institute for Global Orthopedics and Traumatology: A Model for Academic Collaboration in Orthopedic Surgery." *Frontiers in Public Health* 5: 146. https://doi.org/10.3389/fpubh.2017.00146.

Creswell, J. W. 2022. *A Concise Introduction to Mixed Methods Research*. SAGE.

Curran, G. M., S. J. Landes, S. A. McBain, et al. 2022. "Reflections on 10 Years of Effectiveness-Implementation Hybrid Studies." *Frontiers in Health Services* 2: 1053496. https://doi.org/10.3389/frhs.2022.1053496.

Dos Portela Santos, O., P. Melly, R. Hilfiker, et al. 2022. "Effectiveness of Educational Interventions to Increase Skills in Evidence-Based Practice Among Nurses: The EDITcare Systematic Review." *Healthcare (Basel)* 10, no. 11: 2204. https://www.mdpi.com/2227-9032/10/11/2204.

Eldh, A. C., A. Ehrenberg, J. E. Squires, C. A. Estabrooks, and L. Wallin. 2013. "Translating and Testing the Alberta Context Tool for Use Among Nurses in Swedish Elder Care." *BMC Health Services Research* 13: 68. https://doi.org/10.1186/1472-6963-13-68.

Eldh, A. C., E. Joelsson-Alm, P. Wretenberg, and M. Hälleberg-Nyman. 2021. "Onset PrevenTIon of Urinary Retention in Orthopaedic Nursing and Rehabilitation, OPTION—A Study Protocol for a Randomised Trial by a Multi-Professional Facilitator Team and Their First-Line Managers'

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Implementation Strategy." *Implementation Science* 16, no. 1: 65. https://doi.org/10.1186/s13012-021-01135-x.

Elo, S., and H. Kyngäs. 2008. "The Qualitative Content Analysis Process." *Journal of Advanced Nursing* 62, no. 1: 107–115. https://doi.org/10.1111/j.1365-2648.2007.04569.x.

Estabrooks, C. A., J. E. Squires, G. G. Cummings, J. M. Birdsell, and P. G. Norton. 2009. "Development and Assessment of the Alberta Context Tool." *BMC Health Services Research* 9, no. 1: 234. https://doi.org/10.1186/1472-6963-9-234.

Fjordkvist, E., M. Hälleberg Nyman, M. Winberg, E. Joelsson-Alm, and A. C. Eldh. 2024. "First-Line Managers' Experience of Guideline Implementation in Orthopaedic Nursing and Rehabilitation: A Qualitative Study." *BMC Health Services Research* 24, no. 1: 871. https://doi.org/10.1186/s12913-024-11353-w.

Flikweert, E. R., K. W. Wendt, R. L. Diercks, et al. 2018. "Complications After Hip Fracture Surgery: Are They Preventable?" *European Journal of Trauma and Emergency Surgery* 44, no. 4: 573–580. https://doi.org/10.1007/s00068-017-0826-2.

Fry, M., R. Elliott, L. Fitzpatrick, et al. 2020. "Measuring Nurses' Perceptions of Their Work Environment and Linking With Behaviour Change Theories and Implementation Strategies to Support Evidence Based Practice Change." *Applied Nursing Research* 56: 151374. https://doi.org/10.1016/j.apnr.2020.151374.

Gwennan Hâf, B. 2020. *Designing and Conducting Mixed Methods Research*. British Psychological Society.

Harvey, G. 2022. "Context Matters, So How Do we Get Better at Working With Context in Implementation Research and Practice?; Comment on 'Stakeholder Perspectives of Attributes and Features of Context Relevant to Knowledge Translation in Health Settings: A Multi-Country Analysis'." *International Journal of Health Policy and Management* 11, no. 8: 1587–1589.

Harvey, G., and A. Kitson. 2015. "PARIHS Revisited: From Heuristic to Integrated Framework for the Successful Implementation of Knowledge Into Practice." *Implementation Science* 11, no. 1: 1–13.

Harvey, G., and A. Kitson. 2020. *Promoting Action on Research Implementation in Health Services: The Integrated-PARIHS Framework*, 114–143. Edward Elgar Publishing.

International Council of Nurses (ICN). 2021. "The ICN Code of Ethics for Nurses: Revised 2021. International Council of Nurses."

Joelsson-Alm, E., and H. Thulin. 2021. "Blåsövervakning vid Sjukhusvård. [Bladder Monitoring During Hospital Care]."

Kim, C., M. J. Armstrong, W. B. Berta, and A. R. Gagliardi. 2020. "How to Identify, Incorporate and Report Patient Preferences in Clinical Guidelines: A Scoping Review." *Health Expectations* 23, no. 5: 1028–1036. https://doi.org/10.1111/hex.13099.

Lincoln, Y. S., and E. G. Guba. 1986. "But Is It Rigorous? Trustworthiness and Authenticity in Naturalistic Evaluation." *New Directions for Program Evaluation* 1986: 73–84. https://doi.org/10.1002/ev.1427.

Lindseth, A., and A. Norberg. 2004. "A Phenomenological Hermeneutical Method for Researching Lived Experience." *Scandinavian Journal of Caring Sciences* 18, no. 2: 145–153. https://doi.org/10.1111/j.1471-6712.2004.00258.x.

Malterud, K., V. D. Siersma, and A. D. Guassora. 2016. "Sample Size in Qualitative Interview Studies: Guided by Information Power." *Qualitative Health Research* 26, no. 13: 1753–1760. https://doi.org/10.1177/1049732315617444.

Mandal, L., and A. Seethalakshmi. 2023. "Experience of Missed Nursing Care: A Mixed Method Study." *Worldviews on Evidence-Based Nursing* 20, no. 3: 212–219. https://doi.org/10.1111/wvn.12653.

Mathew, D., S. K. Cesario, A. Malecha, and R. Toms. 2024. "Role of Magnet(R)-Recognized Hospital Nurse Managers in Implementing Evidence-Based Practice: A Mixed-Method Study." *Worldviews on Evidence-Based Nursing* 21, no. 1: 23–33. https://doi.org/10.1111/wvn.12693.

Matthew-Maich, N., J. Ploeg, M. Dobbins, and S. Jack. 2013. "Supporting the Uptake of Nursing Guidelines: What You Really Need to Know to Move Nursing Guidelines Into Practice." *Worldviews on Evidence-Based Nursing* 10, no. 2: 104–115.

Melnyk, B. M., and E. Fineout-Overholt. 2022. *Evidence-Based Practice in Nursing & Healthcare: A Guide to Best Practice*. Lippincott Williams & Wilkins.

Moore, G. F., S. Audrey, M. Barker, et al. 2015. "Process Evaluation of Complex Interventions: Medical Research Council Guidance." *BMJ (Clinical Research Ed.)* 350: h1258. https://doi.org/10.1136/bmj.h1258.

Parekh, A. D., J. E. Bates, and R. J. Amdur. 2020. "Response Rate and Nonresponse Bias in Oncology Survey Studies." *American Journal of Clinical Oncology* 43, no. 4: 229–230. https://doi.org/10.1097/COC.0000000000000665.

Pomey, M.-P., D. P. Ghadiri, P. Karazivan, N. Fernandez, and N. Clavel. 2015. "Patients as Partners: A Qualitative Study of Patients' Engagement in Their Health Care." *PLoS One* 10, no. 4: e0122499. https://doi.org/10.1371/journal.pone.0122499.

Santana, M. J., K. Manalili, R. J. Jolley, S. Zelinsky, H. Quan, and M. Lu. 2018. "How to Practice Person-Centred Care: A Conceptual Framework." *Health Expectations* 21, no. 2: 429–440. https://doi.org/10.1111/hex.12640.

Saunders, H., L. Gallagher-Ford, T. Kvist, and K. Vehviläinen-Julkunen. 2019. "Practicing Healthcare Professionals' Evidence-Based Practice Competencies: An Overview of Systematic Reviews." *Worldviews on Evidence-Based Nursing* 16, no. 3: 176–185. https://doi.org/10.1111/wvn.12363.

Squires, J. E., L. Hayduk, A. M. Hutchinson, et al. 2015. "Reliability and Validity of the Alberta Context Tool (ACT) With Professional Nurses: Findings From a Multi-Study Analysis." *PLoS One* 10, no. 6: e0127405. https://doi.org/10.1371/journal.pone.0127405.

Squires, J. E., A. M. Hutchinson, M. Coughlin, et al. 2022. "Stakeholder Perspectives of Attributes and Features of Context Relevant to Knowledge Translation in Health Settings: A Multi-Country Analysis." *International Journal of Health Policy and Management* 11, no. 8: 1373–1390. https://doi.org/10.34172/ijhpm.2021.32.

Thompson, D. S., K. O'Leary, E. Jensen, S. Scott-Findlay, L. O'Brien-Pallas, and C. A. Estabrooks. 2008. "The Relationship Between Busyness and Research Utilization: It Is About Time." *Journal of Clinical Nursing* 17, no. 4: 539–548. https://doi.org/10.1111/j.1365-2702.2007.01981.x.

Toepoel, V., and M. Schonlau. 2017. "Dealing With Nonresponse: Strategies to Increase Participation and Methods for Postsurvey Adjustments." *Mathematical Population Studies* 24, no. 2: 79–83. https://doi.org/10.1080/08898480.2017.1299988.

Tovin, M. M., and M. E. Wormley. 2023. "Systematic Development of Standards for Mixed Methods Reporting in Rehabilitation Health Sciences Research." *Physical Therapy* 103, no. 11: 84. https://doi.org/10.1093/ptj/pzad084.

Translating Research in Elder Care (TREC). 2024. "Alberta Context Tool (ACT) Manual." https://trecresearch.ca/alberta_context_tool.

van der Winden, D., N. van Dijk, M. R. M. Visser, and J. Bont. 2023. "Walking the Line Between Assessment, Improvement and Learning: A Qualitative Study on Opportunities and Risks of Incorporating Peer Discussion of Audit and Feedback Within Quality Improvement in General Practice." *BMJ Open* 13, no. 1: e066793. https://doi.org/10.1136/bmjopen-2022-066793.

Warren, J. I., M. McLaughlin, J. Bardsley, et al. 2016. "The Strengths and Challenges of Implementing EBP in Healthcare Systems." *Worldviews*

13652648, 2026, 4, Downloaded from https://onlinelibrary.wiley.com/doi/10.1111/jan.70098 by Uppsala University Library, Wiley Online Library on [11/06/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

on Evidence-Based Nursing 13, no. 1: 15–24. https://doi.org/10.1111/wvn.12149.

Winberg, M., M. Hälleberg Nyman, E. Fjordkvist, A. C. Eldh, and E. Joelsson-Alm. 2024. "Adherence to Evidence-Based Guidelines for Prevention of Urinary Retention in Hip Surgery Patients: A Multicentre Observational Study." *International Journal for Quality in Health Care* 36, no. 2: 45. https://doi.org/10.1093/intqhc/mzae045.

Winberg, M., M. Hälleberg Nyman, E. Fjordkvist, E. Joelsson-Alm, and A. C. Eldh. 2023. "Patients' Experiences of Urinary Retention and Bladder Care – A Qualitative Study in Orthopaedic Care." *International Journal of Orthopaedic and Trauma Nursing* 50: 101034. https://doi.org/10.1016/j.ijotn.2023.101034.

Younas, A., and A. Durante. 2023. "The Logics of and Strategies to Enhance Generalization of Mixed Methods Research Findings." *Methodology* 19, no. 2: 170–191. https://doi.org/10.5964/meth.10863.

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